February 2026 | 1.30 pm | A-103

Social Networks 2—Social Networks in Cultural and Scientific Fields

Session 7 – Statistical Network Models II

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DEPARTMENT OF NETWORK AND DATA SCIENCE (DNDS)

Session 7 – Statistical network models II

- Models for longitudinal data, stochastic actor-oriented models (SAOMs)

- Methodological background of the models, discussion of link function, estimation procedures etc.

Stochastic actor-oriented models (SAOMs)

- What are the basic ideas behind SAOMs?
- Can you provide an intuition how the models arrive at the parameters?
- What are differences of SAOMs in comparison to ERGMs or other methods you know from your studies?

SAOMs: model architecture

Focal actor aka. ego

Objective function: $$f_i(\beta, x) = \sum_k \beta_k s_{ki}(x).$$

Parameters aka. weights

State of the network

Sum over all effects

Effects, i.e., network structures focal actor *i* can choose from, for instance:

Reciprocity $\sum_j x_{ij}x_{ji},$

transitive triplets $\sum_{j,h} x_{ih}x_{ij}x_{jh},$

Intuition behind obtaining parameters

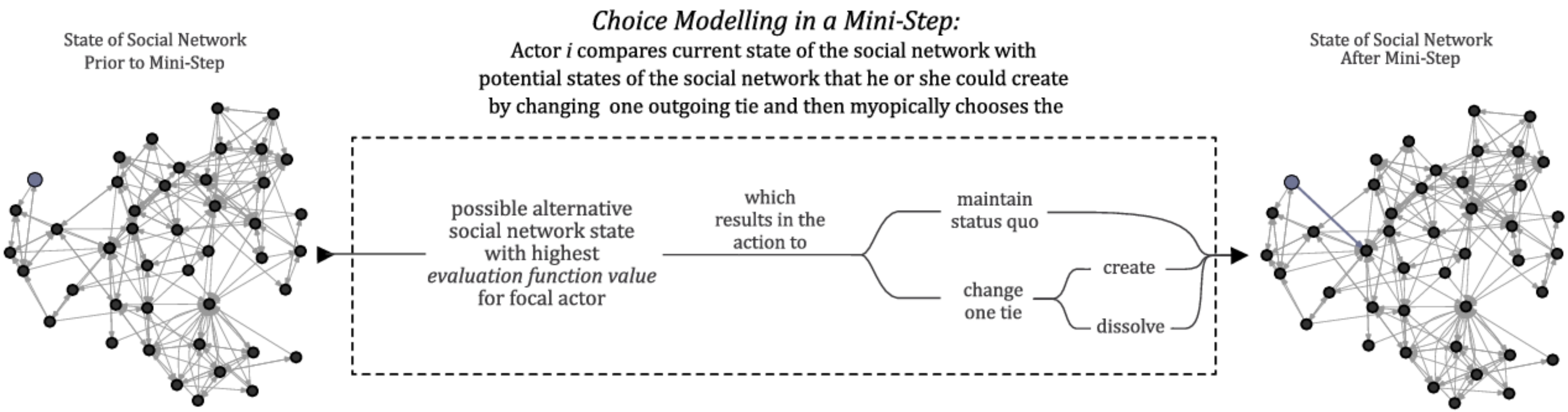
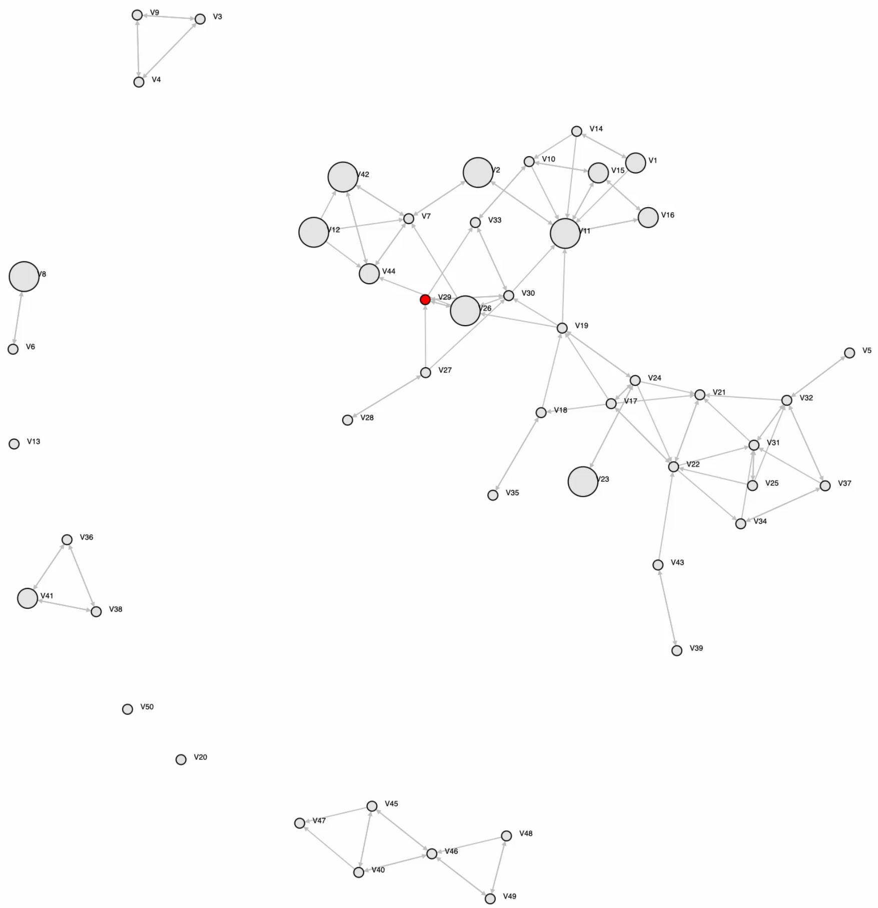


Fig. 1. Illustration of modelling network choices (selection) using the stochastic actor-oriented model.

Pink, S., Kretschmer, D., & Leszczensky, L. (2020). Choice modelling in social networks using stochastic actor-oriented models. *Journal of choice modelling*, *34*,



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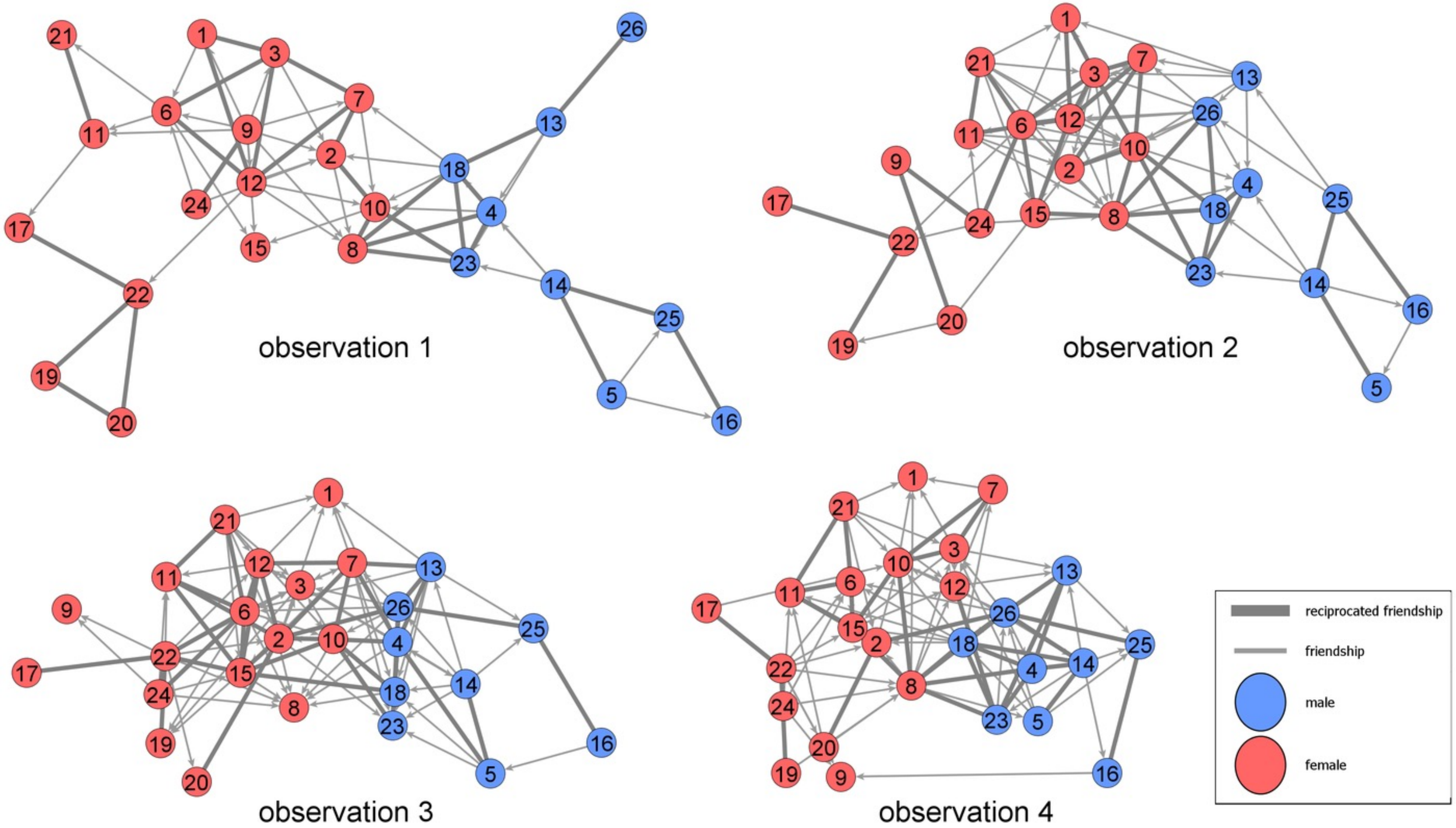
Example: Selection and Influence Processes in School



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Friendship and Delinquency: Selection and Influence Processes in Early Adolescence

Andrea Knecht, *University of Erlangen-Nuremberg*, Tom A. B. Snijders, *University of Oxford and University of Groningen*, Chris Baerveldt, *Utrecht University*, Christian E. G. Steglich, *University of Groningen*, and Werner Raub, *Utrecht University*

Abstract

Positive association of relevant characteristics is a widespread pattern among adolescent friends. A positive association may be caused by the selection of similar others as friends and by the deselection of dissimilar ones, but also by influence processes where friends adjust their behavior to each other. Social control theory argues that adolescents select each other as friends based on delinquency. Differential association theory, on the other hand, argues that adolescent friends influence each other's delinquency levels. We employ new statistical methods for assessing the empirical evidence for either process while controlling for the other process. These methods are based on 'actor-oriented' stochastic simulation models. We analyze longitudinal data on friendship networks and delinquent behavior collected in four waves of 544 students in 21 first-grade classrooms of Dutch secondary schools. Results indicate that adolescents select others as friends who have a similar level of delinquency compared with their own level. Estimates of the social influence parameters are not significant. The results are consistent with social control theory but provide no support for differential association theory.

Table 2. Results of Meta-analysis of SIENA Analyses (21 Classes)

	Estimated mean parameter	*SE*	Estimated true *SD*	*P* value of test that variance of parameter is 0
Selection part				
Shared delinquency level	.160***	.037	<.001	.402
Delinquency ego	–.002	.091	.37	<.001
Delinquency alter	–.077	.044	.15	.019
Outdegree	–1.907***	.027	<.001	.044
Reciprocity	.847***	.053	.16	.011
Transitivity	.194***	.009	.03	.002
Gender similarity	.637***	.081	.32	<.001
Ethnicity similarity	.108	.056	.18	.010
Friends in primary school	.409***	.072	.21	.295
Influence part				
Average delinquency of friends	.032	.156	<.001	.998
Tendency delinquency	–.535***	.071	.16	.045
Tendency delinquency squared	–.000	.062	.19	.092
Male	.387***	.161	.53	.378

* $p < .05$, ** $p < .01$, *** $p < .001$.

Additional examples from Snijders 2010

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	Model 0		
	Estim.	S.E.	*p*
Objective function			
Outdegree	−1.41	0.43	<0.001
Reciprocity (evaluation)	1.34	0.22	<0.001
Reciprocity (endowment)			
Transitive triplets	0.23	0.03	<0.001
Transitive ties	0.74	0.21	<0.001
3-cycles	−0.32	0.10	<0.001
Indegree popularity (sqrt)	−0.18	0.12	0.13
Outdegree popularity (sqrt)	−0.40	0.24	0.093
Outdegree activity (sqrt)	0.01	0.07	0.93
Sex (M) ego	0.35	0.13	0.008
Sex (M) alter	0.10	0.13	0.46
Same sex	0.49	0.13	<0.001
Primary school	0.37	0.13	0.006
Rate function			
Rate period 1	10.01	2.05	
Rate period 2	10.67	1.85	
Rate period 3	9.51	1.53	
Effect sex (M) on rate			

Effect	Estimate	S.E.	*p*
Network objective function			
Outdegree	−1.91	0.41	<0.001
Reciprocity (evaluation)	0.25	0.44	0.57
Reciprocity (endowment)	2.10	0.81	0.010
Transitive triplets	0.22	0.03	<0.001
Transitive ties	0.67	0.23	0.004
3-cycles	−0.27	0.11	0.014
Outdegree based popularity (sqrt)	−0.52	0.25	0.034
Sex (M) ego	0.48	0.16	0.002
Sex (M) alter	0.19	0.16	0.23
Same sex	0.61	0.15	<0.001
Primary school	0.46	0.18	0.010
Delinquency similarity	3.22	1.66	0.053
Network rate function			
Network rate period 1	9.94	2.12	
Network rate period 2	10.86	2.00	
Network rate period 3	9.39	1.49	
Delinquency linear	0.00	0.27	1.00
Delinquency quadratic	0.12	0.16	0.48
Sex (M)	−0.19	0.42	0.66
Average similarity	6.08	3.06	0.047
Behavior rate function			
Delinquency rate period 1	1.50	0.70	
Delinquency rate period 2	3.50	2.48	
Delinquency rate period 3	2.64	1.39	

Introduction to estimation of SAOM parameters

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Overview

1. Define statistical model, relationship between dependent variable and a vector of independent variables

$$\lambda_i(\rho, \alpha, x) = \rho_m \exp\left(\sum_k \alpha_k u_{ik}(x)\right)$$

$$f_i(\beta, x^0, x) = \sum_{k=1}^{L} \beta_k s_{ik}(x^0, x)$$

Rate function and objective function

2. Derivation of estimates for coefficients and standard errors via estimation method, e.g., OLS, ML, MCMC, MoM, etc. → can be frequentist or Bayesian

$$P\{R = r, X_2 = x_2, \ldots, X_{R-1} = x_{r-1}, I_2 = i_2, \ldots, I_R = i_r \mid X(t_1) = x_1, X(t_2) = x_R\}$$
$$= \exp(-n\rho(t_2 - t_1)) \frac{(\rho(t_2 - t_1))^{r-1}}{(r-1)!} \times \prod_{h=2}^{r} \frac{\exp(\sum_k \beta_k s_{i_h k}(x_h))}{\sum_{x \in \mathcal{A}_i(x_{h-1})} \exp(\sum_k \beta_k s_{i_h k}(x))}$$

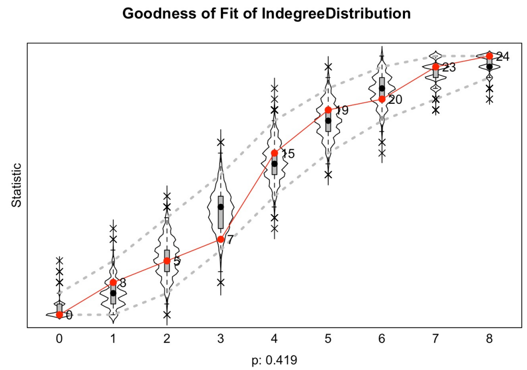
e.g., Maximum likelihood

3. Implementation of estimation procedure in statistical software

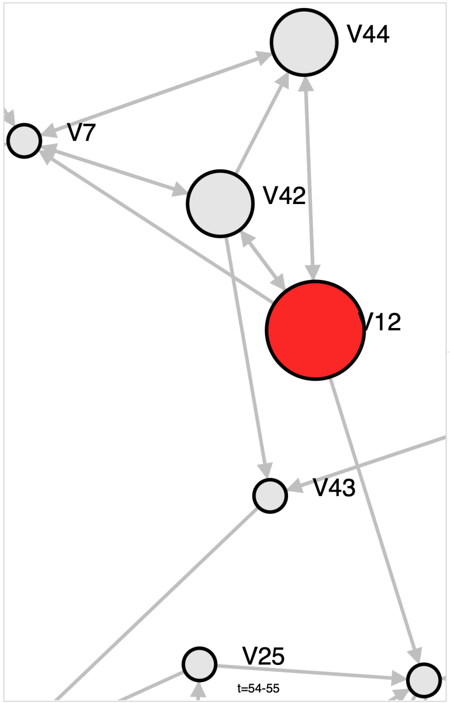
4. Estimation of models with empirical data

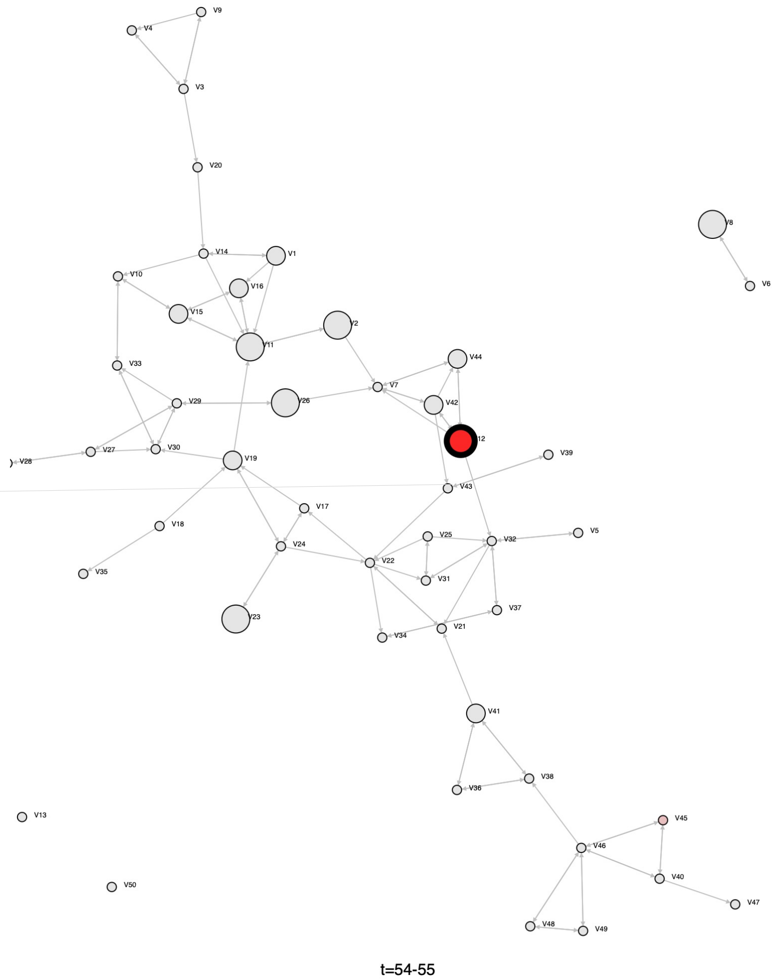
```
## Network Dynamics
##    1. rate constant fr4wav rate (period 1)
##    2. rate constant fr4wav rate (period 2)
##    3. rate constant fr4wav rate (period 3)
##    4. eval outdegree (density)
##    5. eval reciprocity
##    6. eval same gender
##    7. eval smokebeh alter
##    8. eval smokebeh ego
##    9. eval same smokebeh
##
##
##              Estimate   Standard   Convergence
##                           Error      t-ratio
##
##    1.          1.1392  (   0.223  )   -0.0454
##    2.          1.1321  (   0.213  )    0.0056
##    3.          1.1260  (   0.200  )   -0.0086
##    4.         -3.0585  (   0.422  )    0.0695
##    5.          0.8387  (   0.247  )    0.0807
##    6.          1.4113  (   0.303  )    0.0423
##    7.          0.7123  (   0.417  )    0.0127
##    8.         -0.0733  (   0.307  )    0.0721
##    9.          1.2041  (   0.486  )    0.0303
##
```

5. Postestimation: Goodness of fit (GOF), other model diagnostics, additional transformations for interpretation (e.g., relative importance of parameters)



Example selection of a focal actor who can change the network structure





Model architecture of SAOMs in detail: *opportunities* for change

$$\lambda(\alpha, x) = \sum_i \lambda_i(\alpha, x).$$

$$\pi_i(\alpha, x) = \frac{\lambda_i(\alpha, x)}{\lambda(\alpha, x)}.$$

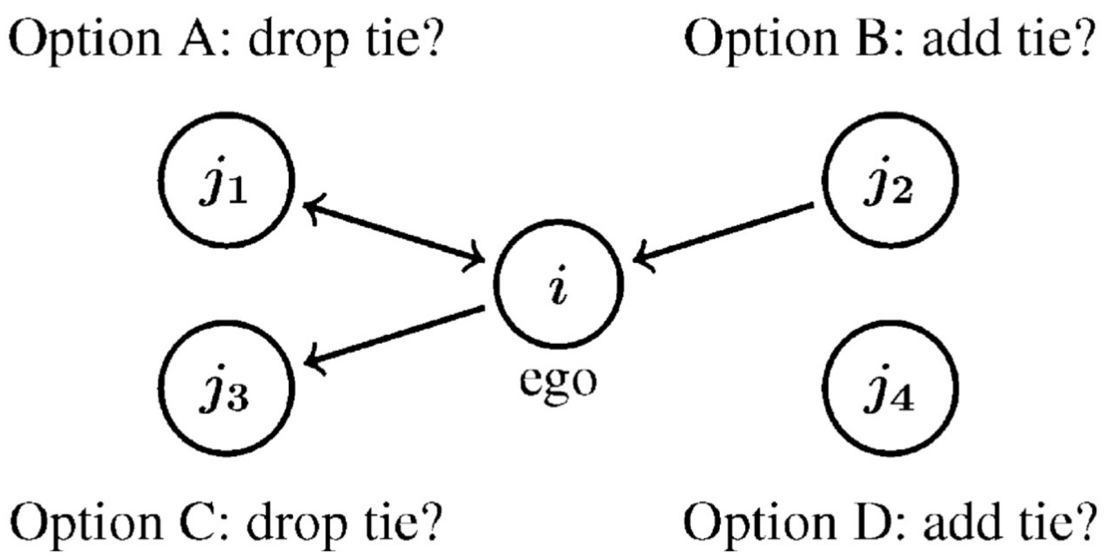
Rate function lambda ist summarizing the parameters in alpha for each actor in the network and determines the opportunities for change (i.e., the waiting time for each actor in the mini-step process); Pi is giving us the probability of actor i to be the focal actor that can make a change

$$\lambda_i(\rho, \alpha, x) = \rho_m \, \exp\left(\sum_k \alpha_k u_{ik}(x)\right),$$

We can make the rate function dependent on positional characteristics of actors, expressed by u_{ik} between arbitrary timesteps captured by rho

For details see: Snijders, T. A. (2017). Stochastic actor-oriented models for network dynamics. ***Annual review of statistics and its application***, ***4***(1), 343-363.

Example for a focal actor i determining which *options* to choose in a mini-step



Model architechture of SAOMs in detail: *options* for change

$$f_i(\beta, x^0, x) = \sum_{k=1}^{L} \beta_k \, s_{ik}(x^0, x),$$

Objective function given parameters beta_k for network effects s_{ik}

$$s_{ik}(x^0, x) = \sum_j x_{ij} \qquad \text{(outdegree)},$$

$$\sum_j x_{ij}\, x_{ji} \qquad \text{(reciprocated ties)},$$

$$\sum_{j,k} x_{ij}\, x_{jk}\, x_{ik} \qquad \text{(transitive triplets), and}$$

$$\sum_j x^0_{ij}\, x^0_{ji}\, x_{ij}\, x_{ji} \qquad \text{(persistent reciprocity).}$$

A small selection of potential effects that can be included in the objective function → analysts choice: theory + data-driven assessment which effects work and matter

$$p_i(\beta, x^0, x) = \begin{cases} \dfrac{\exp(f_i(\beta, x^0, x))}{\sum_{\tilde{x}} \exp(f_i(\beta, x^0, \tilde{x}))} & x \in \mathcal{A}_i(x^0) \\ 0 & x \notin \mathcal{A}_i(x^0), \end{cases}$$

The conditional probability that the next digraph is x, given that the current digraph is x_0 and actor i gets the opportunity to make a change, is assumed to be given by the multinomial model

For details see: Snijders, T. A. (2017). Stochastic actor-oriented models for network dynamics. ***Annual review of statistics and its application***, ***4***(1), 343-363.

Methods of moments estimation for SAOMs I

We want to estimate this vector of parameters theta:

$$\theta = (\rho, \alpha, \beta)$$

$$D(X(t_{m+1}), X(t_m)) = \sum_{i,j} \mid X_{ij}(t_m + 1) - X_{ij}(t_m) \mid .$$

$$A_k(X(t_{m+1}), X(t_m)) = \sum_{i,j} u_{ik}(X(t_m)) \mid X_{ij}(t_m + 1) - X_{ij}(t_m) \mid .$$

$$s_k(X(t_m)) = \sum_{i} s_{ik}(X(t_m)).$$

For each type of parameters we define an one-dimensional statistic that is senstive to the respective parameters:

1. For rho (the total amount of change) we use the Hamming distance between network states.
2. For alpha (determining how strongly the rate of change for actor i is influenced by ui k (X)), we use the second statistic.
3. For beta (network effects), the third statistic is used, which is sensitive to sik

For details see: Snijders, T. A. (2017). Stochastic actor-oriented models for network dynamics. ***Annual review of statistics and its application***, ***4***(1), 343-363.

Methods of moments estimation for SAOMs II

Combining these statistics and employing the Markov chain assumption for the observed data $x(t_1), \ldots, x(t_M)$, the estimating equations are

$$D(x(t_{m+1}), x(t_m)) = E_\theta\{D(X(t_{m+1}), x(t_m)) \mid X(t_m) = x(t_m)\} \quad (m = 1, \ldots, M-1), \tag{15a}$$

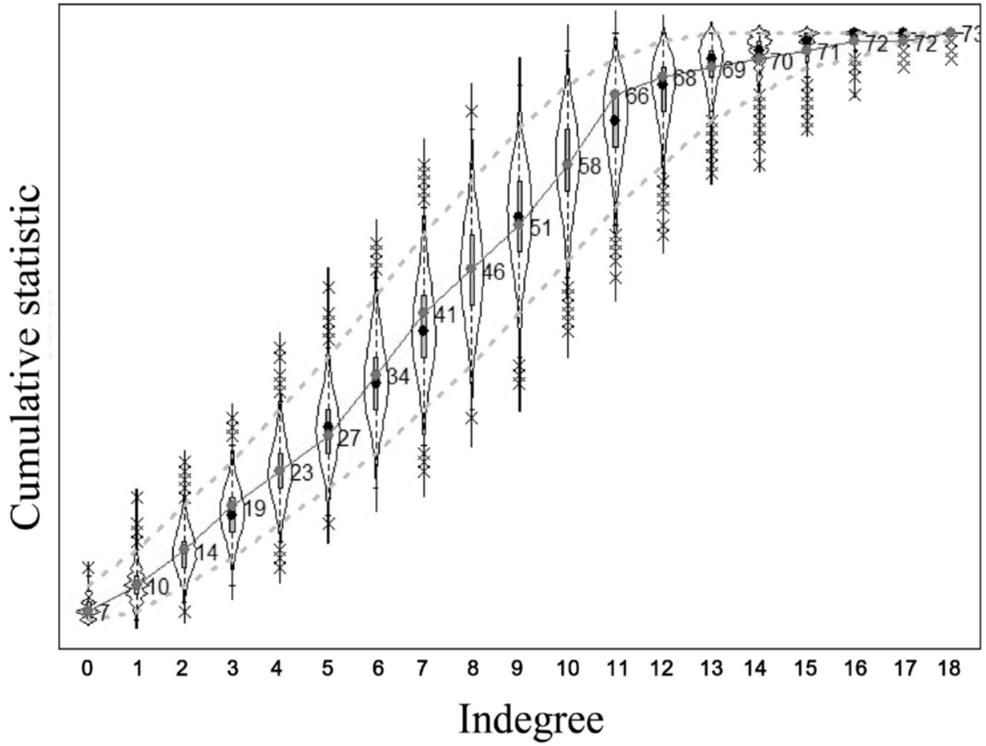
$$\sum_{m=1}^{M-1} A_k(x(t_{m+1}), x(t_m)) = \sum_{m=1}^{M-1} E_\theta\{A_k(X(t_{m+1}), x(t_m)) \mid X(t_m) = x(t_m)\} \quad (k = 1, \ldots, K_\alpha), \tag{15b}$$

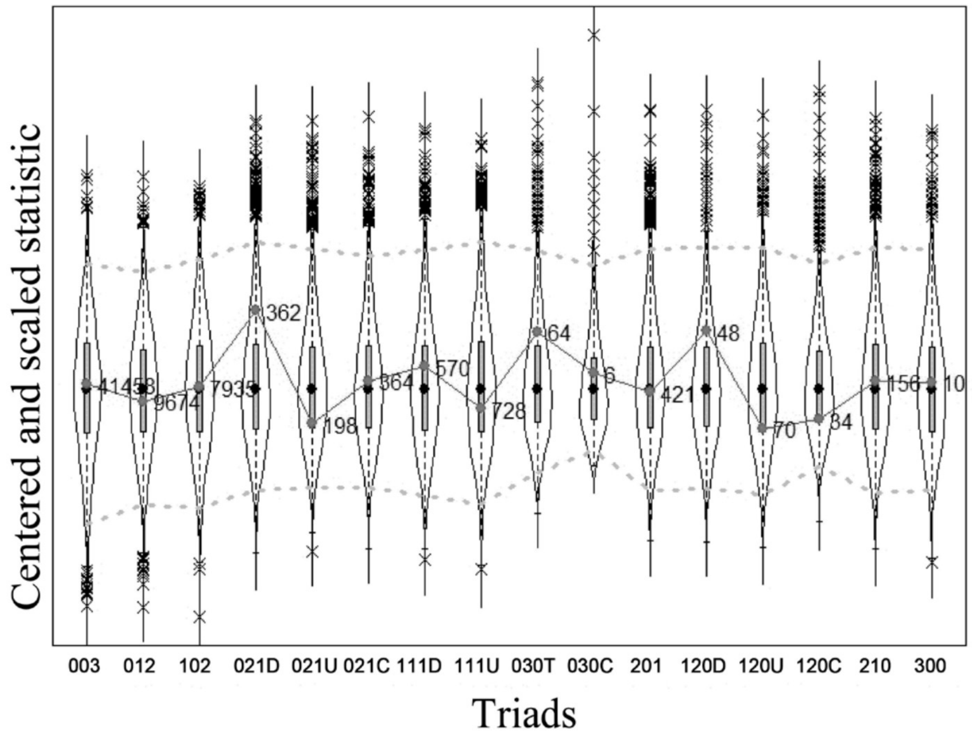
$$\sum_{m=1}^{M-1} s_k(x(t_{m+1})) = \sum_{m=1}^{M-1} E_\theta\{s_k(X(t_{m+1})) \mid X(t_m) = x(t_m)\} \quad (k = 1, \ldots, K_\beta), \tag{15c}$$

Can be solved with Robbins-Monro algorithm, see Snijders, T. A. (2001). The statistical evaluation of social network dynamics. ***Sociological methodology***, ***31***(1), 361-395.

For details see: Snijders, T. A. (2017). Stochastic actor-oriented models for network dynamics. ***Annual review of statistics and its application***, ***4***(1), 343-363.

Model diagnostics: Goodness of fit (GOF) example





Some model extensions

- SAOMs for co-evolution of networks and behavior
- Multiplex SAOMs for co-evolution of multiple types of ties
- Bi-partite SAOMs
- Bayesian SAOMs, especially for pooling many networks in one model, increasing usage over the last decade

Summary

- SAOMs allow to model the evolution of networks by considering how focal actors form, delete, or maintain ties
- Interpretation of parameters similar to logOdds in logistic regression models
- Typical way to obtain parameters is method of moments or maximum likelihood
- Goodness of fit is assessed by simulating networks from an estimated SAOM and checking whether network properties of simulated networks are similar to empirical network

Organizational details

- Last two weeks of the term will be dedicated to student presentations, each group should prepare a **~15 minutes presentation**, slots will be distributed next week before we start the workshop part.

- Next week will be half discussion, half lab session. Bring your **laptops** and pre-install **Rsiena**

- If you want to dive deeper into SAOMs, see the slides from Koskinen and Snijders that I uploaded in Moodle and read the additional texts on technical foundation of the models

Readings for next week

Lewis, K., & Kaufman, J. (2018). The conversion of cultural tastes into social network ties. ***American journal of sociology***, ***123***(6), 1684-1742.

Try to get the basic Rsiena script running on your laptop:
https://www.stats.ox.ac.uk/~snijders/siena/

Thank you for your attention!